

Unique Machine Learning Algorithms

Deduplicated reference — 18 unique algorithms

1. Linear Regression

```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(X_train, y_train)
y_pred = regressor.predict(X_test)
```

2. Logistic Regression

```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression(random_state=0)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

3. Ridge Regression

```
from sklearn.linear_model import Ridge
regressor = Ridge(alpha=1.0)
regressor.fit(X_train, y_train)
y_pred = regressor.predict(X_test)
```

4. Lasso Regression

```
from sklearn.linear_model import Lasso
regressor = Lasso(alpha=0.1)
regressor.fit(X_train, y_train)
y_pred = regressor.predict(X_test)
```

5. Elastic Net

```
from sklearn.linear_model import ElasticNet
regressor = ElasticNet(alpha=0.1, l1_ratio=0.5)
regressor.fit(X_train, y_train)
y_pred = regressor.predict(X_test)
```

6. Decision Tree

```
from sklearn.tree import DecisionTreeClassifier
classifier = DecisionTreeClassifier(criterion='entropy', random_state=0)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

7. Random Forest

```
from sklearn.ensemble import RandomForestClassifier
classifier = RandomForestClassifier(n_estimators=200, criterion='entropy', random_state=0)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

8. Gradient Boosting

```
from sklearn.ensemble import GradientBoostingClassifier
classifier = GradientBoostingClassifier()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

9. XGBoost

```
from xgboost import XGBClassifier
classifier = XGBClassifier()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

10. LightGBM

```
from lightgbm import LGBMClassifier
classifier = LGBMClassifier()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

11. CatBoost

```
from catboost import CatBoostClassifier
classifier = CatBoostClassifier()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

12. AdaBoost

```
from sklearn.ensemble import AdaBoostClassifier
classifier = AdaBoostClassifier()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

13. k-Nearest Neighbors

```
from sklearn.neighbors import KNeighborsClassifier
classifier = KNeighborsClassifier(n_neighbors=5)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

14. Naive Bayes

```
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

15. Support Vector Machine

```
# kernel can be swapped: 'rbf', 'linear', 'poly', etc.
from sklearn.svm import SVC
classifier = SVC(kernel='rbf', probability=True, random_state=0)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

16. Linear Discriminant Analysis (LDA)

```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
classifier = LinearDiscriminantAnalysis()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

17. Quadratic Discriminant Analysis (QDA)

```
from sklearn.discriminant_analysis import QuadraticDiscriminantAnalysis
classifier = QuadraticDiscriminantAnalysis()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```

18. Multilayer Perceptron (Neural Network)

```
from sklearn.neural_network import MLPClassifier
classifier = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state=42)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
```